Ayasha Hossain Jui

Software Engineer | Backend & FinTech
+880 1521-410529 | ayashajui1110@gmail.com
linkedin.com/in/ayasha-hossain-jui | github.com/ayashaJui | leetcode.com/u/ayashaJui | ayashaJui.github.io

Summary

Backend software engineer with 3+ years of experience developing securities trading platforms in **Java and Spring Boot**. Focused on correctness-critical systems: financial transaction processing, **concurrency-safe order and balance handling**, and real-time market data pipelines on Kafka and RabbitMQ. Also develops Angular front ends and has published research on deep-learning image classification.

Experience

Software Engineer March 2023 – Present
Dohatec New Media – Independent Software Vendor *Dhaka, Bangladesh*

- Develop and maintain production microservices in **Java and Spring Boot** for PQCal, a live securities trading platform. For PQCal Global (in development), own the backend end to end across user management, the financial module, configuration, file handling, and authentication, along with the supporting infrastructure (service registry, config server, API gateway).
- Engineered the **financial module** for an international trading product, covering deposits, withdrawals, and instrument trading, with real-time balance updates delivered to the Angular UI over Server-Sent Events (SSE).
- Prevented duplicate orders and conflicting balance updates under concurrent or retried requests by making balance and margin updates atomic, using database transactions, **pessimistic row locking**, and **idempotency keys**.
- Contributed to the market-data service, which consumes live CME (Chicago Mercantile Exchange) data from **Kafka** and produces market depth, option chain, candlestick, and time-and-sales views served to the Angular UI through REST APIs.
- Developed versioned REST APIs for user management, authentication, financial operations, and market analysis.
- Optimized SQL Server and PostgreSQL queries using Spring Data JPA and Hibernate, diagnosing slow queries and API performance issues in financial-data workflows.
- Established asynchronous, event-driven communication between services using **RabbitMQ and Kafka**, decoupling them and improving reliability.
- Designed authentication and authorization on Spring Security, JWT, and OAuth 2.0 as a **standalone authorization server**, isolated from other services so users can still sign in when any one of them is down; extended it to support international expansion.
- Implemented reusable **Angular** components with PrimeNG and secure token-based authentication flows.
- Delivered changes through Docker-based deployments and CI/CD pipelines with QA and DevOps; wrote unit and functional tests and took part in design reviews, code reviews, and Agile sprint planning.

Projects

PQCal Global (*Under Development*) Dohatec New Media
Java, Spring Boot, Angular, Kafka, RabbitMQ, SQL Server, PostgreSQL

International version of PQCal for markets outside Bangladesh, with live paper trading on CME market data.

- Owned the backend for accounts, payments and trading (deposits, withdrawals, instrument orders), configuration, file handling, and authentication, plus the service registry, config server, and API gateway.
- Contributed to the market-data service: market depth, option chain, candlestick, and time-and-sales views from the CME feed on Kafka.
- Developed Angular features across the web application.

PQCal (*Live Production*) pqcal.dohasecurities.com.bd
Spring Boot, FastAPI, Angular, PrimeNG, SQL Server, PostgreSQL, RabbitMQ, Docker

Dohatec's first in-house product: a live securities trading platform used in the Bangladeshi market.

- Developed user-management REST APIs in Spring Boot with search, pagination, and cleanup of incomplete sign-ups, using RabbitMQ events to keep services in sync.
- Implemented market-analysis APIs in Python and FastAPI that serve company fundamentals, dividend history, index data, and intraday charts.
- Delivered the user registration flow (a multi-step form with document uploads) and portfolio reporting screens in Angular and PrimeNG.

Dohatec AI (*RAG Application*) Dohatec New Media
Angular, Spring Boot RAG backend (team-developed), LLM

Company AI assistant, developed by a cross-functional team, that answers questions from users' own documents through a chat interface.

- Developed the Angular document-management and chat features that connect users to the retrieval-augmented backend.
- Analyzed and helped resolve backend issues across the **RAG pipeline** (document ingestion, chunking, embeddings, semantic retrieval, and LLM answer generation) in close collaboration with backend developers.

Biblioteca (*Personal Project*) github.com/ayashaJui/book_tracking

Java 21, Spring Cloud, OAuth 2.0/JWT, PostgreSQL, Angular, RxJS, Docker

Book-tracking platform with a catalogue of books, authors, publishers, and series, plus personal libraries and reading preferences, designed as independent microservices.

- Designed separate Spring Boot services for users, catalogue, and personal libraries, each with its own database, behind **Spring Cloud Gateway** with **Eureka** service discovery and centralized configuration.
- Developed a dedicated **OAuth 2.0 authorization server** that issues JWTs, which resource servers validate; the Angular app signs in through the Authorization Code flow.
- Managed schema changes with Flyway, documented APIs with OpenAPI/Swagger, and containerized core services with Docker Compose.
- Implemented the **Angular 20** front end with PrimeNG, Tailwind CSS, and RxJS across 14 lazy-loaded feature modules, with an HTTP interceptor that attaches the access token.

StudyNest (*Live*) studynest-z0r3.onrender.com

Next.js, NestJS, TypeScript, MongoDB, JWT, Swagger, GitHub Actions

Interview-prep platform with nested topics, Markdown notes, flashcards, quizzes, and a coding-problem tracker.

- Developed a full-stack application with a Next.js front end and a separate **NestJS REST API** backed by MongoDB; migrated backend functionality from Next.js API routes and deployed the two services on Render.
- Implemented an **SM-2-based spaced-repetition system** for flashcards and interval-based scheduling for coding-problem reviews, with a daily review queue, timezone-aware study streaks, and global search.
- Secured the application with HttpOnly cookie-based JWT authentication, bcrypt password hashing, and email password reset; added CSV and Markdown import and export.

Skin Cancer Classification (*Research Project*) IEEE ICCIT 2022

Python, TensorFlow, Pandas, NumPy, CNN, Xception, Transfer Learning, ISIC 2019

Deep-learning approach to classifying skin cancer from images, published at the 25th International Conference on Computer and Information Technology (ICCIT).

- Fine-tuned the pre-trained **Xception CNN** using **transfer learning** to classify **eight types of skin cancer**, applying data augmentation to increase training-data variety.
- Evaluated the model on the ISIC 2019 dataset, reporting better results than comparable studies.

Ink & Paper inkandpaper-frontend.onrender.com

React, Redux, Node.js, Express.js, MongoDB, Material UI

Full-stack e-commerce bookstore.

- Developed user authentication, book browsing, author and review exploration, and a shopping cart.
- Designed REST APIs with Express and MongoDB, and a responsive React UI with Material UI and Redux for global state management.

TourVille (*University Project*) github.com/ayashaJui/CSEKU_WPL_2020_tourville

PHP, MySQL, JavaScript, Bootstrap, jQuery, Stripe

Tourism platform for discovering, booking, and managing travel packages, developed with a teammate for the Web Programming Lab at Khulna University.

- Developed a multi-role system for tourists, travel agencies, and administrators, including package search, agency profiles, bookings with Budget, Comfortable, and Luxury pricing tiers, reviews, and admin account approval.
- Integrated **Stripe payments**, allowing tourists to pay in full, in half, or through a percentage deposit.
- Implemented role-based agency accounts (Owner, Manager, Staff) and an analytics dashboard with Google Charts; secured data access with PDO prepared statements and bcrypt-hashed passwords.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, PHP, C++, SQL

Problem Solving: 350+ LeetCode problems solved in Java, Python, and C++ (arrays, strings, hash tables, binary search, dynamic programming, backtracking)

Backend: Spring Boot, Spring Cloud (Gateway, Eureka, Config), Spring MVC, Spring Security, Spring Data JPA/Hibernate, Flyway, FastAPI, Node.js, Express.js, NestJS

Frontend: Angular (NgRx, PrimeNG, Tailwind CSS), React (Redux, Material UI), Next.js, HTML, CSS, jQuery

AI & GenAI: Retrieval-Augmented Generation (RAG), LLM integration, document embeddings, semantic search, text chunking

Machine Learning: TensorFlow, Pandas, NumPy, Convolutional Neural Networks (CNNs), transfer learning (Xception), data augmentation, image classification

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB

Messaging & Streaming: Apache Kafka, RabbitMQ, Server-Sent Events (SSE)

DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD, Swagger/OpenAPI

Practices: Microservices, event-driven architecture, REST API design and versioning, concurrency-safe transactions, idempotency, JWT/OAuth 2.0, OOP and design patterns, clean code, unit testing (JUnit, Mockito), code reviews, Agile/Scrum

Education

Khulna University Khulna, Bangladesh
B.Sc. Engineering in Computer Science and Engineering — CGPA: 3.72/4.00 *2018 – 2023*

Publications

Ayasha Hossain Jui, Samia Sharnami, Aminul Islam. *"A CNN Based Approach to Classify Skin Cancers using Transfer Learning"*, 25th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 17–19 December 2022. DOI: 10.1109/ICCIT57492.2022.10055838